We create chemistry

Safety data sheet

1. Substance/preparation and manufacturer/supplier identification

Opal 7.5

Use: crop protection product, fungicide

Manufacturer/supplier:
BASF (Thai) Limited
23rd Floor, Emporium Tower, 622, Sukhumvit 24 Rd.,
Klongton, Klongtoey, Bangkok 10110, THAILAND
Telephone: +66 2624-1999
Telefax number: +66 2664-9254
E-mail address: Thailand-SDS-info@basf.com

Emergency information:
International emergency number:
Telephone: +49 180 2273-112

2. Hazard identification

Classification according to UN GHS 2009

Classification of the substance and mixture:
Aspiration hazard: Cat. 1
Flammable liquids: Cat. 4
Acute toxicity: Cat. 4 (oral)
Skin corrosion/irritation: Cat. 3
Serious eye damage/eye irritation: Cat. 2A
Carcinogenicity: Cat. 2
Reproductive toxicity: Cat. 2 (fertility)
Reproductive toxicity: Cat. 2 (unborn child)
Specific target organ toxicity — single exposure: Cat. 3 (irritating to respiratory system)
Hazardous to the aquatic environment - acute: Cat. 2
Hazardous to the aquatic environment - chronic: Cat. 2

Label elements and precautionary statement:

Pictogram:

Signal Word:
Danger

Hazard Statement:

H227	Combustible liquid.
H319	Causes serious eye irritation.
H316	Causes mild skin irritation.
H302	Harmful if swallowed.
H304	May be fatal if swallowed and enters airways.
H335	May cause respiratory irritation.
H351	Suspected of causing cancer.
H361	Suspected of damaging fertility. Suspected of damaging the unborn child.
H401	Toxic to aquatic life.
H411	Toxic to aquatic life with long lasting effects.

Precautionary Statement:

P101	If medical advice is needed, have product container or label at hand.
P102	Keep out of reach of children.
P103	Read carefully and follow all instructions.

Precautionary Statements (Prevention):

P280	Wear protective gloves, protective clothing and eye protection or face protection.
P201	Obtain special instructions before use.
P271	Use only outdoors or in a well-ventilated area.
P210	Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking.
P202	Do not handle until all safety precautions have been read and understood.
P264	Wash contaminated body parts thoroughly after handling.
P261	Avoid breathing mist or vapour or spray.
P270	Do not eat, drink or smoke when using this product.

Precautionary Statements (Response):

P312	Call a POISON CENTER or physician if you feel unwell.
P305 + P351 + P338	IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.
P301 + P310	IF SWALLOWED: Immediately call a POISON CENTER or physician.
P304 + P340	IF INHALED: Remove person to fresh air and keep comfortable for breathing.
P308 + P313	IF exposed or concerned: Get medical attention.
P330	Rinse mouth
P391	Collect spillage.
P331	Do NOT induce vomiting.
P337 + P313	If eye irritation persists: Get medical attention.
P332 + P313	If skin irritation occurs: Get medical attention.
P370 + P378	In case of fire: Use water spray, dry powder, foam or carbon dioxide for extinction.

Precautionary Statements (Storage):

P403 + P233	Store in a well-ventilated place. Keep container tightly closed.
P405	Store locked up.

Precautionary Statements (Disposal):

P501	Dispose of contents and container to hazardous or special waste collection point.

Other hazards which do not result in classification:
See section 12 - Results of PBT and vPvB assessment.
If applicable information is provided in this section on other hazards which do not result in classification but which may contribute to the overall hazards of the substance or mixture.

3. Composition/information on ingredients

Chemical nature

crop protection product, fungicide, Emulsifiable concentrate (EC)

Hazardous ingredients

Epoxiconazole

Content (W/W): 7.34 %	Acute Tox.: Cat. 5 (oral)
CAS Number: 133855-98-8	Carc.: Cat. 2
	Repr.: Cat. 2 (fertility)
	Repr.: Cat. 2 (unborn child)
	Aquatic Acute: Cat. 1
	Aquatic Chronic: Cat. 1
	M-factor acute: 10
	M-factor chronic: 10

benzyl alcohol

Content (W/W): < 30 %	Acute Tox.: Cat. 4 (oral)
CAS Number: 100-51-6	Acute Tox.: Cat. 4 (Inhalation - mist)
	Acute Tox.: Cat. 4 (dermal)
	Eye Dam./Irrit.: Cat. 2A
	Aquatic Acute: Cat. 2

solvent naphtha

Content (W/W): < 30 %	Asp. Tox.: Cat. 1
CAS Number: 64742-94-5	Carc.: Cat. 2
	Aquatic Acute: Cat. 2
	Aquatic Chronic: Cat. 2

Naphthalene, 2-methyl-

Content (W/W): < 15 %	Acute Tox.: Cat. 4 (oral)
CAS Number: 91-57-6	Aquatic Chronic: Cat. 2

Poly(oxy-1,2-ethanediyl), .alpha.-tridecyl-.omega.-hydroxy-, branched

Content (W/W): < 15 %	Eye Dam./Irrit.: Cat. 2B
CAS Number: 69011-36-5	Aquatic Acute: Cat. 2
	Aquatic Chronic: Cat. 3

naphthalene

Content (W/W): < 10 %	Flam. Sol.: Cat. 2
CAS Number: 91-20-3	Acute Tox.: Cat. 4 (oral)
	Carc.: Cat. 2
	Aquatic Acute: Cat. 1
	Aquatic Chronic: Cat. 1
	M-factor acute: 1
	M-factor chronic: 1

Naphthalene, 1-methyl-

Content (W/W): < 10 %	Asp. Tox.: Cat. 1
CAS Number: 90-12-0	Acute Tox.: Cat. 4 (oral)
	Aquatic Acute: Cat. 2
	Aquatic Chronic: Cat. 2

sodium-di-ethyl-hexyl-sulfosuccinate

Content (W/W): < 5 %	Acute Tox.: Cat. 5 (oral)
CAS Number: 577-11-7	Skin Corr./Irrit.: Cat. 2
	Eye Dam./Irrit.: Cat. 1
	Aquatic Acute: Cat. 2

Hydrocarbons, C10, aromatics, <1% naphthalene

Content (W/W): < 3 % CAS Number: 64742-94-5	Asp. Tox.: Cat. 1 Flam. Liq.: Cat. 4 STOT SE: Cat. 3 (drowsiness and dizziness) Aquatic Acute: Cat. 2 Aquatic Chronic: Cat. 2

4. First-Aid Measures

General advice:
First aid personnel should pay attention to their own safety. If the patient is likely to become unconscious, place and transport in stable sideways position (recovery position). Immediately remove contaminated clothing.

If inhaled:
Keep patient calm, remove to fresh air, seek medical attention.

On skin contact:
Immediately wash thoroughly with soap and water, seek medical attention.

On contact with eyes:
Wash affected eyes for at least 15 minutes under running water with eyelids held open, consult an eye specialist.

On ingestion:
Immediately rinse mouth and then drink 200-300 ml of water, seek medical attention. Do not induce vomiting due to aspiration hazard.

Note to physician:
Symptoms: Information, i.e. additional information on symptoms and effects may be included in the GHS labeling phrases available in Section 2 and in the Toxicological assessments available in Section 11., (Further) symptoms and / or effects are not known so far
Treatment: Treat according to symptoms (decontamination, vital functions), no known specific antidote.

5. Fire-Fighting Measures

Suitable extinguishing media:
water spray, foam, dry powder, carbon dioxide

Unsuitable extinguishing media for safety reasons:
water jet

Specific hazards:
carbon monoxide, hydrogen chloride, hydrogen fluoride, carbon dioxide, nitrogen oxides, sulfur oxides, organochloric compounds, silicon oxides
The substances/groups of substances mentioned can be released in case of fire.

Special protective equipment:
Wear self-contained breathing apparatus and chemical-protective clothing.

Further information:
Collect contaminated extinguishing water separately, do not allow to reach sewage or effluent systems. Dispose of fire debris and contaminated extinguishing water in accordance with official regulations. In case of fire and/or explosion do not breathe fumes. Keep containers cool by spraying with water if exposed to fire.

6. Accidental Release Measures

Personal precautions:
Do not breathe vapour/spray. Use personal protective clothing. Avoid contact with the skin, eyes and clothing.

Environmental precautions:
Do not discharge into drains/surface waters/groundwater. Do not discharge into the subsoil/soil.

Methods for cleaning up or taking up:
For small amounts: Pick up with suitable absorbent material (e.g. sand, sawdust, general-purpose binder, kieselguhr).
For large amounts: Dike spillage. Pump off product.
Dispose of absorbed material in accordance with regulations. Collect waste in suitable containers, which can be labeled and sealed. Clean contaminated floors and objects thoroughly with water and detergents, observing environmental regulations. Wear suitable protective equipment.

7. Handling and Storage

Handling

No special measures necessary if stored and handled correctly. Ensure thorough ventilation of stores and work areas. When using do not eat, drink or smoke. Hands and/or face should be washed before breaks and at the end of the shift.

Protection against fire and explosion:
Vapours may form ignitable mixture with air. Prevent electrostatic charge - sources of ignition should be kept well clear - fire extinguishers should be kept handy.

Storage

Segregate from foods and animal feeds.
Further information on storage conditions: Keep away from heat. Protect from direct sunlight.

Storage stability:
Storage duration: 60 Months

8. Exposure controls and personal protection

Components with occupational exposure limits

naphthalene, 91-20-3;

TWA value 10 ppm (ACGIHTLV)
Skin Designation (ACGIHTLV)
Danger of cutaneous absorption
Skin Designation (ACGIHTLV)
Danger of cutaneous absorption

solvent naphtha, 64742-94-5;
TWA value 200 mg/m3 (ACGIHTLV), Non-aerosol
Measured as: total hydrocarbon vapor
Application restricted to conditions in which there are negligible aerosol exposures.
Skin Designation (ACGIHTLV), Non-aerosol
Measured as: total hydrocarbon vapor
Danger of cutaneous absorption

epoxyconazole, 133855-98-8;
TWA value 0.3 mg/m3 (Recommendation of BASF), Respirable dust

Naphthalene, 1-methyl-, 90-12-0;
TWA value 0.5 ppm (ACGIHTLV)
Skin Designation (ACGIHTLV)
Danger of cutaneous absorption

Naphthalene, 2-methyl-, 91-57-6;
TWA value 0.5 ppm (ACGIHTLV)
Skin Designation (ACGIHTLV)
The substance can be absorbed through the skin.
Skin Designation (ACGIHTLV)
Danger of cutaneous absorption

Solvent naphtha (petroleum), heavy arom., 64742-94-5;
TWA value 200 mg/m3 (ACGIHTLV), Non-aerosol
Measured as: total hydrocarbon vapor
Application restricted to conditions in which there are negligible aerosol exposures.
Skin Designation (ACGIHTLV), Non-aerosol
Measured as: total hydrocarbon vapor
Danger of cutaneous absorption

Personal protective equipment

Respiratory protection:
Suitable respiratory protection for lower concentrations or short-term effect: Combination filter for gases/vapours of organic, inorganic, acid inorganic, alkaline compounds and toxic particles (e. g. EN 14387 Type ABEK-P3)

Hand protection:
Suitable chemical resistant safety gloves (EN ISO 374-1) also with prolonged, direct contact (Recommended: Protective index 6, corresponding > 480 minutes of permeation time according to EN ISO 374-1): E.g. nitrile rubber (0.4 mm), chloroprene rubber (0.5 mm), butyl rubber (0.7 mm) etc.

Eye protection:
Safety glasses with side-shields (frame goggles) (e.g. EN 166)

Body protection:
Body protection must be chosen depending on activity and possible exposure, e.g. apron, protecting boots, chemical-protection suit (according to EN 14605 in case of splashes or EN ISO 13982 in case of dust).

General safety and hygiene measures:
The statements on personal protective equipment in the instructions for use apply when handling crop-protection agents in final-consumer packing. Wearing of closed work clothing is recommended. Store work clothing separately. Keep away from food, drink and animal feeding stuffs.

9. Physical and Chemical Properties

Form:	liquid	
Colour:	yellow	
Odour:	moderate odour, aromatic	
Odour threshold:	Not determined due to potential health hazard by inhalation.	
pH value:	approx. 5.5 - 7.5 (1.0 %(m), 20 °C) (as an emulsion)	
crystallization temperature:	< -20 °C	
onset of boiling:	approx. 200 °C Information applies to the solvent.	
Flash point:	93 °C	(Directive 92/69/EEC, A.9)
Evaporation rate:	not applicable	
Flammability (solid/gas):	not applicable	
Lower explosion limit:	As a result of our experience with this product and our knowledge of its composition we do not expect any hazard as long as the product is used appropriately and in accordance with the intended use.	
Upper explosion limit:	As a result of our experience with this product and our knowledge of its composition we do not expect any hazard as long as the product is used appropriately and in accordance with the intended use.	
Ignition temperature:	420 °C	(Directive 92/69/EEC, A.15)
Thermal decomposition:	70 °C , 20 kJ/kg	(DSC (DIN 51007))
	240 °C , 20 kJ/kg	(DSC (DIN 51007))

	325 °C , 120 kJ/kg (onset temperature) Not a substance liable to self-decomposition according to UN transport regulations, class 4.1.	(DSC (DIN 51007))
Explosion hazard:	Based on the chemical structure there is no indication of explosive properties.	
Fire promoting properties:	Based on its structural properties the product is not classified as oxidizing.	
Vapour pressure:	approx. 0.005 hPa (20 °C) Information applies to the solvent.	
Density:	approx. 1.02 g/cm3 (20 °C)	(OECD Guideline 109)
Relative vapour density (air):	not applicable	
Solubility in water:	emulsifiable	
Partitioning coefficient n-octanol/water (log Pow):	not applicable	
Viscosity, dynamic:	approx. 10 mPa.s (20 °C, 100 1/s)	

Other Information:
If necessary, information on other physical and chemical parameters is indicated in this section.

10. Stability and Reactivity

Conditions to avoid:
See SDS section 7 - Handling and storage.

Thermal decomposition:	70 °C, 20 kJ/kg (DSC (DIN 51007))
Thermal decomposition:	240 °C, 20 kJ/kg (DSC (DIN 51007))
Thermal decomposition:	325 °C, 120 kJ/kg (DSC (DIN 51007)) (onset temperature) Not a substance liable to self-decomposition according to UN transport regulations, class 4.1.

Substances to avoid:
strong oxidizing agents, strong acids, strong bases

Hazardous reactions:
No hazardous reactions if stored and handled as prescribed/indicated.

Hazardous decomposition products:
No hazardous decomposition products if stored and handled as prescribed/indicated.

11. Toxicological Information

Acute toxicity

Assessment of acute toxicity:
Of moderate toxicity after single ingestion. Virtually nontoxic by inhalation. Virtually nontoxic after a single skin contact.

Experimental/calculated data:
LD50 rat (oral): > 500 - < 2,000 mg/kg (OECD Guideline 423)

LC50 rat (by inhalation): > 5.4 mg/l 4 h (OECD Guideline 403)
An aerosol was tested.

LD50 rat (dermal): > 4,000 mg/kg (OECD Guideline 402)
No mortality was observed.

Irritation

Assessment of irritating effects:
Eye contact causes irritation. Skin contact causes slight irritation.

Experimental/calculated data:
Skin corrosion/irritation rabbit: (OECD Guideline 404)

Serious eye damage/irritation rabbit: (OECD Guideline 405)

Respiratory/Skin sensitization

Assessment of sensitization:
There is no evidence of a skin-sensitizing potential.

Experimental/calculated data:
modified Buehler test guinea pig: (OECD Guideline 406)

Germ cell mutagenicity

Assessment of mutagenicity:
The product has not been tested. The statement has been derived from the properties of the individual components.

Information on: naphthalene
Assessment of mutagenicity:
The substance was not mutagenic in bacteria. The substance was mutagenic in a mammalian cell culture test system. The substance was not mutagenic in a test with mammals. Literature data.

Carcinogenicity

Assessment of carcinogenicity:
The product has not been tested. The statement has been derived from the properties of the individual components.

Information on: Epoxiconazole
Assessment of carcinogenicity:
Indication of possible carcinogenic effect in animal tests.

Information on: solvent naphtha
Assessment of carcinogenicity:
Indication of possible carcinogenic effect in animal tests.

Information on: naphthalene
Assessment of carcinogenicity:
In long-term studies in rats and mice in which the substance was given by inhalation, a carcinogenic effect was observed. EU-classification The substance was classified as a group 3 carcinogen by the German MAK-Commission (substances for which a suspicion of a carcinogenic potential exists). IARC (International Agency for Research on Cancer) has classified this substance as group 2B (The agent is possibly carcinogenic to humans).

Reproductive toxicity

Assessment of reproduction toxicity:
The product has not been tested. The statement has been derived from the properties of the individual components.

Information on: Epoxiconazole
Assessment of reproduction toxicity:
The results of animal studies suggest a fertility impairing effect.

Developmental toxicity

Assessment of teratogenicity:
The product has not been tested. The statement has been derived from the properties of the individual components.

Information on: Epoxiconazole
Assessment of teratogenicity:
Indications of possible developmental toxicity/teratogenicity were seen in animal studies.

Information on: sodium-di-ethyl-hexyl-sulfosuccinate
Assessment of teratogenicity:
The substance did not cause malformations in animal studies; however, toxicity to development was observed at high doses which impaired body weight gain in parental animals.

Specific target organ toxicity (single exposure):

Assessment of STOT single:
Causes temporary irritation of the respiratory tract.

Remarks: The product has not been tested. The statement has been derived from the properties of the individual components.

Repeated dose toxicity and Specific target organ toxicity (repeated exposure)

Assessment of repeated dose toxicity:
The product has not been tested. The statement has been derived from the properties of the individual components.

Information on: Epoxiconazole
Assessment of repeated dose toxicity:
Repeated exposure to large quantities may affect certain organs.

Information on: naphthalene
Assessment of repeated dose toxicity:
The substance may cause damage to the olfactory epithelium after repeated inhalation.

Aspiration hazard

May also damage the lung at swallowing (aspiration hazard).
The product has not been tested. The statement has been derived from the properties of the individual components.

Other relevant toxicity information

Misuse can be harmful to health.

Has a degreasing effect on skin.

12. Ecological Information

Ecotoxicity

Assessment of aquatic toxicity:
Toxic to aquatic life with long lasting effects.

Toxicity to fish:
LC50 (96 h) approx. 4.3 mg/l, Lepomis macrochirus (OECD Guideline 203, static)

Aquatic invertebrates:
EC50 (48 h) 6.49 mg/l, Daphnia magna (OECD Guideline 202, part 1, static)

Aquatic plants:
EC50 (72 h) > 10 mg/l (growth rate), Pseudokirchneriella subcapitata (OECD Guideline 201)

Mobility

Assessment transport between environmental compartments:
The product has not been tested. The statement has been derived from the properties of the individual components.

Information on: Epoxiconazole
Assessment transport between environmental compartments:
Following exposure to soil, adsorption to solid soil particles is probable, therefore contamination of groundwater is not expected.

Persistence and degradability

Assessment biodegradation and elimination (H2O):
The product has not been tested. The statement has been derived from the properties of the individual components.

Information on: Epoxiconazole
Assessment biodegradation and elimination (H2O):
Not readily biodegradable (by OECD criteria).

Bioaccumulation potential

Assessment bioaccumulation potential:
The product has not been tested. The statement has been derived from the properties of the individual components.

Information on: Epoxiconazole
Bioaccumulation potential:
Bioconcentration factor: 59 - 70, Oncorhynchus mykiss (OECD-Guideline 305)
Does not accumulate in organisms.

Additional information

Other ecotoxicological advice:
Do not discharge product into the environment without control.

13. Disposal Considerations

Must be sent to a suitable incineration plant, observing local regulations.

Contaminated packaging:
Contaminated packaging should be emptied as far as possible and disposed of in the same manner as the substance/product.

14. Transport Information

Domestic transport:

Packing group:	III
ID number:	UN 3082
Transport hazard class(es):	9, EHSM
Proper shipping name:	ENVIRONMENTALLY HAZARDOUS SUBSTANCE, LIQUID, N.O.S. (contains EPOXICONAZOLE, NAPHTHALENE)

Sea transport
IMDG

Packing group:	III

ID number:	UN 3082
Transport hazard class(es):	9, EHSM
Marine pollutant:	YES
Proper shipping name:	ENVIRONMENTALLY HAZARDOUS SUBSTANCE, LIQUID, N.O.S. (contains EPOXICONAZOLE, NAPHTHALENE)

Air transport
IATA/ICAO

Packing group:	III
ID number:	UN 3082
Transport hazard class(es):	9, EHSM
Proper shipping name:	ENVIRONMENTALLY HAZARDOUS SUBSTANCE, LIQUID, N.O.S. (contains EPOXICONAZOLE, NAPHTHALENE)

Further information
Product may be shipped as non-hazardous in suitable packages containing a net quantity of 5 L or less under the provisions of various regulatory agencies: ADR, RID, ADN: Special Provision 375; IMDG: 2.10.2.7; IATA: A197; TDG: Special Provision 99(2); 49CFR: §171.4 (c) (2).

15. Regulatory Information

Hazard determining component(s) for labelling: EPOXICONAZOLE, BENZYLALCOHOL, SOLVENT NAPHTHA

Other regulations

To avoid risks to man and the environment, comply with the instructions for use.

16. Other Information

Vertical lines in the left hand margin indicate an amendment from the previous version.

The data contained in this safety data sheet are based on our current knowledge and experience and describe the product only with regard to safety requirements. This safety data sheet is neither a Certificate of Analysis (CoA) nor technical data sheet and shall not be mistaken for a specification agreement. Identified uses in this safety data sheet do neither represent an agreement on the corresponding contractual quality of the substance/mixture nor a contractually designated use. It is the responsibility of the recipient of the product to ensure any proprietary rights and existing laws and legislation are observed.